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The Actual result, Pass/Fail, and Initials/date columns are blank on purpose. Any simulated run must be labeled SIMULATED.

1. **Purpose**

   This protocol tests that Steam Sterilizer S-101 and its control system operate as specified in Part 1 across their operating range, including failure and challenge conditions. It covers every High and Medium risk requirement in the FMEA (Part 2) that is tested in OQ, plus the severity-3 door lock.

2. **Scope and Coverage**

| Test | Title | Traces To |
|---|---|---|
| OQ-01 | Access Control | URS-3.1 |
| OQ-02 | Password, Lockout, and inactivity logout | URS-3.2,3.3,3.4 |
| OQ-03 | Audit trail and record protection | URS-8.2, 8.3 |
| OQ-04 | Electronic signature and cycle release | URS-2.1, URS-4.4 |
| OQ-05 | Calibration status check | URS-4.1 |
| OQ-06 | Empty-Chamber temperature mapping | URS-6.1,6.2,6.3 |
| OQ-07 | Air removal test | URS-6.3 |
| OQ-08 | Vacuum leak test | URS-6.4 |
| OQ-09 | Low-temperature alarm challenge | URS-6.4, URS-9.2 |
| OQ-10 | Failed Cycle, deviation, and release block | URS-2.4 |
| OQ-11 | Power failure challenge | URS-7.3 |
| OQ-12 | Backup and restore | URS-7.1,7.2 URS-8.3 |
| OQ-13 | Door interlock | URS-9.1 |
| OQ-14 | Operator authorization and load record | URS-2.3,2.4 |
| OQ-15 | Requalification control and recipe approval | URS-2.2, URS-4.3 |
| OQ-16 | Time synchronization and clock protection | URS-8.4, URS-5.1 |
| OQ-17 | Data recording and report content | URS-8.1, URS-4.2 |

   Tested elsewhere: URS-6.3 (F0) and loaded temperature mapping in PA; URS-6.4 (steam quality) in a separate third-party report; URS -1.1 and URS-9.4 in IQ.

3. **Prerequisites**

   - The IQ is approved and complete.
   - This protocol is approved by QA before execution.
   - Testers are trained on S-101 and on this protocol.
   - Test equipment is calibrated, and its ID and calibration due date are recorded below.

| Test equipment | ID | Calibration |
|---|---|---|
| Reference thermocouples, at least 12, with data logger | | |
| Calibrated pressure gauge | | |
| Air removal (Bowie-Dick-type) test pack | | |
| Supplier-approved test leak or calibrated orifice | | |
| Supplier-approved temperature simulator for TT-101 | | |
| Stopwatch | | |

Test accounts needed: One each of Operator, Supervisor, AQ Reviewer, Administrator, Service Engineer, Validation Engineer, and Validation Manager.

## 4. Execution rules

1. Execute steps in order. Write the actual result for every step, even when it matches the expected result.
2. User blue or black in. Date and initial every entry. Correct a mistake with a single line, the date, initials, and a reason.
3. Attach printouts, screenshots, and probe files, labeled with the test and step number.
4. If an actual result differs from the expected result, mark the step Fail, stop, and raise a deviation. Do not repeat the step until QA approves how to proceed.
5. Never overwrite or discard a result.

## 5. Test case

### OQ-01 Access Control

**Traces to**: URS-3.1, FRS-3.1.1,3.1.2,3.1.3.
**Objective**: Show that only a supervisor with a unique ID can change cycle parameters, and that the change needs reason and password.

| Step | Action | Expected result | Actual result | Pass/Fail | Initials/Date |
|---|---|---|---|---|---|
| 1 | Log in as an operator and try to change the exposure time. | The change is refused | | | |

| | | | | | |
|---|---|---|---|---|---|
| 2 | Log in as a supervisor and change the exposure time to test value. | The system asks for a reason and the user's password before saving. | | | |
| 3 | Enter the reason and the correct password. | The change is saved. | | | |
| 4 | Set the exposure time back to 20 minutes, with a reason. | The change is Saved. | | | |
| 5 | Check the audit trail for steps 3 and 4. | Two entries show user ID, date and time, parameter, old value, new value, and reason | | | |

**OQ-02 Passwords, lockout, and inactivity logout**

**Traces to**: URS-3.2,3.3,3.4; FRS-3.2.1,3.3.1,3.3.2,3.4.1.
**Objective**: Show that password rules, account lockout, and inactivity logout work.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | As Administrator, set a test user's password to 7 characters. | Rejected, with the failed rule stated. | | | |
| 2 | Set a password with 8 characters but no special character. | Rejected, with the failed rule stated. | | | |
| 3 | Set a password with 8 | Accepted. | | | |

| | | | | | |
|---|---|---|---|---|---|
| | characters that meets all rules. | | | | |
| 4 | Enter a wrong password for the test user 5 times in a row | The account is locked after the 5th attempt | | | |
| 5 | Check the audit trail. | The lockout is recorded | | | |
| 6 | Try to unlock the account as a supervisor. | Refused. | | | |
| 7 | Unlock the account as Administrator | The account is unlocked and the unlock is recorded in the audit trial. | | | |
| 8 | Log in as a supervisor and start a cycle (or a hold state). Leave the HMI idle. | The screen dims at 3 minutes, and the user is logged out at 5 minutes. The cycle keeps running. | | | |

**OQ-3 Audit trail and record protection**

**Traces to:** URS-8.2,8.3 FRS-8.2.1,8.2.2 8.3.1,8.3.2.
**Objective:** Show that changes are recorded and that either the audit trail nor completed cycle records can be altered.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | As Supervisor, change a critical parameter without entering a reason. | The system will not save the change until a reason is entered. | | | |
| 2 | Enter a reason and save. | The audit trail shows user ID, date and time, what changed, old value, new | | | |

| | | | | | |
|---|---|---|---|---|---|
| | | value, and reason. | | | |
| 3 | As Administrator, try to edit and then delete an audit trail entry. | Both attempts are refused. | | | |
| 4 | Complete a cycle (or open a completed cycle record). Try to edit, delete or overwrite it as a supervisor and as Administrator. | All attempts are refused. | | | |
| 5 | Check the audit trail for the attempts in steps 3 and 4. | The attempts are recorded where the system logs them. | | | |

**OQ-4 Electronic signature and cycle release**

**Traces to:** URS-4.3, URS-2.1; FRS-2.1.1, 2.1.2,2.1.3, FRS-4.4.1, 4.4.2.
**Objective:** Show that only a QA Reviewer can sign a cycle record, and that the signature is complete and cannot be changed.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Complete a cycle and open its record. | The status is pending review | | | |
| 2 | Try to apply for the reviewed signature as an Operator. | Refused | | | |
| 3 | As a QA Reviewer, apply the signature | Refused; the record stays pending review | | | |

| | | | | | |
|---|---|---|---|---|---|
| | with a wrong password. | | | | |
| 4 | Apply the signature with the correct password. | The status changes to Reviewed. | | | |
| 5 | Inspect the signature on screen and on the printout. | It shows printed name, date and time, meaning (Reviewed), and record ID. | | | |
| 6 | As Administrator, try to edit, move, or remove the signature | Refused. | | | |

**OQ-05 Calibration Status Check**

**Traces to:** URS-4.1; FRS-4.1.1, 4.1.2,4.1.3,4.1.4.
**Objective**: Show that a cycle cannot start when an instrument calibration is overdue.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Open the instrument status screen. | The calibration due dates of TT-101 and PT-101 are shown. | | | |
| 2 | As Service Engineer, set the TT-101 due date to a past date (test entry with certificate reference) | The entry is saved and recorded in the audit trail. | | | |
| 3 | As an Operator, try to start a cycle. | The start is blocked and the message Calibration | | | |

| | | | | | |
|---|---|---|---|---|---|
| | | overdue is shown. | | | |
| 4 | As Service Engineer, restore the correct due date. | The entry is saved and recorded in the audit trail. | | | |
| 5 | As an Operator, try to start a cycle. | The cycle starts. | | | |

**OQ-06 Empty-chamber temperature mapping**

**Traces to:** URS-6.1, 6.2,6.3, FRS-6.1.1,6.1.2,6.1.3,6.1.4,6.2.1 ,6.3.2.
**Objective**: Show that the chamber reaches and holds temperature uniformly with no load. Repeat for 3 consecutive cycles.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Place at least 12 reference thermocouples through the validation port at the positions in the mapping diagram. Run cycle C-121 with an empty chamber. | The cycle starts and the probes record. | | | |
| 2 | Watch exposure timer. | The timer starts only when TT-101 reads 121.0C or higher. | | | |
| 3 | During exposure, read TT-101 at the start, middle, and end. | TT-101 Stays between 121.0C and 123.0C. | | | |

| | | | | | |
|---|---|---|---|---|---|
| 4 | At the end of the cycle, find each probe's minimum and maximum during exposure. | Each probe stays between 121.0C and 124.0 C. | | | |
| 5 | Find the largest difference between any two probes at the same time during exposure. | The difference is 2.0 C or less. If it exceeds 1.5 C, the alert was shown. | | | |
| 6 | Record the total cycle time from start to end. | The cycle completes within 90 minutes. | | | |
| 7 | Repeat steps 1 to 6 for 2 more consecutive cycles. | All 3 cycles meet steps 2 to 6. | | | |

**OQ-07 Air removal test**

**Traces to:** URS-6.4, FRS-6.4.1,6.4.2.
**Objective:** Show that air is removed before heat-up and that the air removal test cycle works.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Plan a test pack in the chamber and run the air removal test cycle. | The cycle runs and records its result. | | | |
| 2 | Remove the test pack and inspect it. | The indicator shows a uniform color change, with no pattern of trapped air. | | | |

| | | | | | |
|---|---|---|---|---|---|
| 3 | Check the system's recorded result. | The system records pass. | | | |
| 4 | In the data of C-121 cycle, count the vacuum pulses before heat -up. | At least 3 pulses, each reaching 70mbar absolute or lower. | | | |

**OQ-08 Vacuum leak test**

**Traces to:** URS-6.1, FRS-6.1.3.
**Objective:** Show that the leak test measures the leak rate correctly and fails a leaking chamber.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Run the automatic leak test on the sealed chamber. | The chamber is evacuated and isolated, and pressure is recorded for 10 minutes. | | | |
| 2 | Read the reported leak rate and result. | The rate is 1.3 mbar per minute or less and the result is pass. | | | |
| 3 | Introduce the supplier-approved test leak and run the leak test again. | The rate exceeds 1.3 mbar per minute and the result is Fail. | | | |
| 4 | Remove the test leak and run the test again | The result returns to Pass. | | | |

**OQ- 09 Low-temperature alarm challenge**

**Traces to:** URS-6.3, FRS-6.3.1
**Objective:** Show that a temperature drop during exposure raises and alarm and fails the cycle.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Start a cycle. When exposure begins, use the supplier-approved simulator to make TT-101 read below 121,0 C. | An audible and a visual alarm occur within 2 seconds. Time the alarm with stopwatch. | | | |
| 2 | Read the alarm message. | It shows an alarm ID and the cause in plain English with no raw PLC codes. | | | |
| 3 | Let the cycle finish and read the status on the display and the cycle report. | The cycle is marked Failed on both and is never shown as passed. | | | |
| 4 | Check that the alarm was recorded. | The alarm is recorded with date and time. | | | |

**OQ-10 Failed Cycle, deviation, and release block**

**Traces to:** URS-2.5, FRS-2.5.1
**Objective:** Show that a failed cycle creates a deviation and cannot be released until the deviation is closed. Use the failed cycle from OQ-09.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Look in the quality system (DevTriage in this project) for a new deviation. | A deviation exists with the cycle ID and the failure reason. | | | |

| | | | | | |
|---|---|---|---|---|---|
| 2 | As a QA Reviewer, try to apply the Reviewed signature to the failed cycle. | Refused when deviation is open. | | | |
| 3 | Close the deviation in the quality system. | The deviation status is closed. | | | |
| 4 | As a QA Reviewer, apply for the Reviewed signature again. | The signature is accepted. | | | |

**OQ-11 Power Failure challenge**

**Traces to:** URS-7.3; FRS-7.3.1,7.3.2.
**Objective:** Show that a power loss during a cycle leaves the sterilizer in a safe state with its records kept. Use only the supplier- approved method for cutting power.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Start a cycle. During exposure, cut the main power. | The UPS keeps the PLC, HMI and instruments running . Start the stopwatch. | | | |
| 2 | Confirm the steam inlet valves and the vent valve. | The steam inlet valves close. The vent valve opens or stays controlled so pressure does not rise. | | | |
| 3 | Try to open the door while the chamber is above | The door stays locked | | | |

| | | | | | |
|---|---|---|---|---|---|
| | atmospheric pressure. | | | | |
| 4 | Measure how long the UPS keeps the system running. | At least 10 minutes. | | | |
| 5 | Restore Power | The system starts in a recovery state and requires a supervisor acknowledgement before the next cycle. | | | |
| 6 | Open the interrupted cycle record. | The record written before the failure are present, and the cycle is marked FAILED. | | | |

**OQ-12 Backup and Restore**

**Traces to:** URS-7.1,7.2 ; FRS-7.1.1,7.1.2,7.2.1;
**Objective:** Show that backup run and verify, that a failed backup alerts users, and that data can be restored.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Run the automatic 24-hour backup ran and that cycle records were copied to the secondary location. | A backup file exists on the separate storage location with a verified checksum, and the secondary copy is present. | | | |
| 2 | Disconnect the backup storage and trigger a backup. | A high-priority alarm appears on the HMI within 2 seconds, and the failure is | | | |

| | | | | | |
|---|---|---|---|---|---|
| | | recorded in the audit trail | | | |
| 3 | Reconnect the storage and run backup. | The backup completes and verifies. | | | |
| 4 | As administrator, restore the backup to a test environment. Record the start and end times. | The restore completes within 24 hours. | | | |
| 5 | Compare the restored cycle records and audit trail with the originals. | They match, with the same checksums. | | | |
| 6 | Read the restore report and the audit trail. | The report shows date, backup file used, user ID, and verification result, and the restore is in the audit trail. | | | |

**OQ-13 Door Interlock**

**Traces to:** URS-9.1; FRS:9.1.1
**Objective:** Show that the door cannot be opened while the chamber is pressurized. This is a safety test: follow the supplier's safety instructions.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | During a cycle, when chamber pressure is above 0.05bar, try to open the door from the HMI. | The chamber is evacuated and isolated, and pressure is recorded for 10 minutes. | The door stays locked. | | |

| | | | | | |
|---|---|---|---|---|---|
| 2 | At the same pressure, try to unlatch the door by hand. | The rate is 1.3 mbar per minute or less and the result is pass. | The mechanical lock prevents unlatching. | | |
| 3 | When the cycle ends and chamber pressure returns to atmospheric, open the door. | The rate exceeds 1.3 mbar per minute and the result is Fail. | The door opens. | | |

**OQ-14 Operator authorization and load record**

**Trace to:** USR-2.3,2.4; FSR-2.3.1,2.3.2,2.4.1,2.4.2

**Objective:** Show that only trained operators can start a cycle and that each cycle is linked to its load record.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | As Administrator, confirm and Operator account has no trained-on-s-101 flag. Log in as that Operator and try to start a cycle. | The start is blocked and the blocked attempt is in the audit trail. | | | |
| 2 | As Administrator, set the trained flag for that Operator. | The flag is set and the change is in the audit trail. | | | |
| 3 | As the Operator, try to start a cycle without | The start is blocked. | | | |

| | | | | | |
|---|---|---|---|---|---|
| | selecting a load configuration. | | | | |
| 4 | Select load configuration LC-01 and start. | The cycle starts. | | | |
| 5 | Open the cycle record and try to edit the load configuration, operator ID or date. | The data are shown and cannot be edited. | | | |
| 6 | Print the cycle report | It shows the load configuration, operator name, and date. | | | |

**OQ-15 Requalification control and recipe approval**

**Traces to:** USR-2.2; FSR-2.2.1,2.2.2, 2.2.3, 2.2.4, 2.2.5

**Objective:** Show that an overdue requalification or a changed recipe blocks production, and that approval needs two signatures.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Try to set the requalification due date as a supervisor. | Refused: Only the validation Manger can change it. | | | |
| 2 | As validation Manager, set the due date 25 days ahead. | The system shows a warning that requalification is due within 30 days. | | | |
| 3 | Set the due date a past date (test entry). | The status changes to Requalification Due and | | | |

| | | | | | |
|---|---|---|---|---|---|
| | | production cycles are blocked with the message Execution blocked, sterilizer requires requalification. | | | |
| 4 | Try to enter validation run mode without a change control number, then with a validation engineer account and a number and reason. | Refused without the number. Allowed with it, the cycle is tagged with the number. | | | |
| 5 | As Supervisor, change the parameter of recipe C-121. | The recipe state changes to Unvalidated and production cycles with it are blocked. | | | |
| 6 | Approve the recipe with only the validation manager signature. | The recipe is not approved. | | | |
| 7 | Add the QA reviewer signature. | The recipe becomes Approved for production, and the audit trail records a configuration checksum. | | | |
| 8 | Restore the status Qualified with only one signature, then | Refused with one; Qualified only after both. | | | |

| | | | | | |
|---|---|---|---|---|---|
| | with both signatures. | | | | |

**OQ-16 Time synchronization and clock protection**

**Traces to:** URS-5.1, URS-8.4; FRS-5.1.1,5.1.2, FRS-8.4.1

**Objective:** Show that the system clock follows the site time server and cannot be changed by users.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Compare the HMI time with the site time server. | The times match within 1 second (example tolerance). | | | |
| 2 | As Administrator, try to change the system date or time from the HMI. | Refused. | | | |
| 3 | During a running cycle, trigger a time server adjustment using the supplier-approved method. | The adjustment is held until the cycle ends. | | | |
| 4 | After the cycle ends, check the system time and the audit trail. | The time is corrected and the adjustment is recorded in the audit trail. | | | |

**OQ- 17 Data recording and report content**

**Traces to:** URS-8.1, URS-4.2; FRS-8.1.1, FRS-4.2.1,4.2.2,4.2.3

**Objective:** Show that cycle data are recorded without gaps and that reports are complete.

| Step | Action | Expected Result | Actual Result | Pass/Fail | Initials/date |
|---|---|---|---|---|---|
| 1 | Run a cycle, export the recorded data. | Time, TT-101 temperature, and PT-101 pressure are recorded at least every 5 seconds from start to end, with no gaps. | | | |
| 2 | Compare the report on screen with the printed report. | Same layout, showing cycle ID, batch number, recipe name, operator name, date, and time. | | | |
| 3 | Export the cycle report as PDF. | The file is a read-only PDF/A and its checksum is recorded in the audit trail. | | | |
| 4 | Recalculate the checksum of exported file. | It matches the checksum in the audit trail. | | | |

## 6. Deviation log

| Deviation ID | Test and step | Description | Impact | Resolution | QA approval |
|---|---|---|---|---|---|
| | | | | | |

## 7. Test Summary

| Test | Result (Pass/ Fail) | Deviations | Tester initials/ date |
|---|---|---|---|
| OQ-1 | | | |
| OQ-2 | | | |

| | | | |
|---|---|---|---|
| OQ-3 | | | |
| OQ-4 | | | |
| OQ-5 | | | |
| OQ-6 | | | |
| OQ-7 | | | |
| OQ-8 | | | |
| OQ-9 | | | |
| OQ-10 | | | |
| OQ-11 | | | |
| OQ-12 | | | |
| OQ-13 | | | |
| OQ-14 | | | |
| OQ-15 | | | |
| OQ-16 | | | |
| OQ-17 | | | |

Conclusion (completed after execution) : OQ passes only if every step of every test passes and no critical or major deviation remains open.

## 8. Approvals

| Role | Name | Signature | date |
|---|---|---|---|
| Prepared by (Validation Engineer) | | | |
| Reviewed by (validation Manager) | | | |
| Approved by (QA) before execution | | | |
| Executed by | | | |
| Reviewed by (QA) after execution | | | |